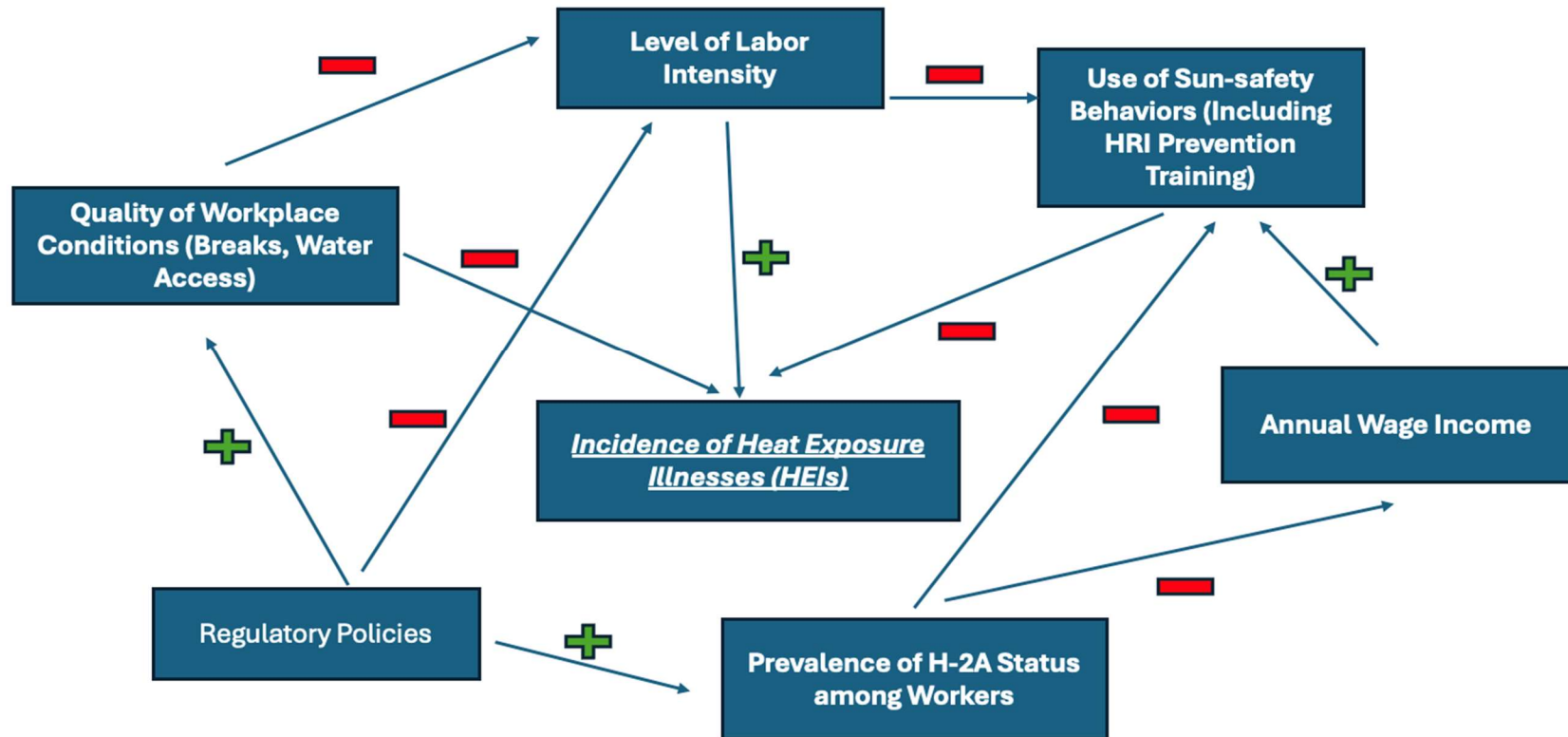


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Systems Map:



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Leverage Point Table: Heat Exposure illnesses among farm and agricultural workers in the rural, Southern United States

Determinant Name + Description	SEF Level	System Location	Importance	Changeability Potential for Leverage
Level of Workforce Intensity Level of workforce intensity refers to the workload and duration of work an individual engages in, particularly in physical labor such as harvesting and loading crops.¹ The level of intensity is measured through prescreening procedures, including phone or in-person interviews with participants or community organizations about their workload and work hours.²	Individual Level This determinant operates at the individual level of the socioecological framework (SEF). According to the operationalization, workforce intensity reflects the physical demands of farmwork on the individual rather than on a community or group.¹ Studies have shown that the nature of work varies across individuals and their physical capabilities.^{2,3}	Incidence of Heat Exposure Illnesses (HEIs) (downstream): Studies assessing the association of heat stress, using core body temperature, and workload intensity found that there was a 12% increase in the odds of reaching or exceeding the physiological limit of 38 C (100.5 F).¹⁻³ Quality of Work Conditions (downstream): A risk factor related to labor management and intensity is employer control over shade availability and proximity to water facilities.¹ Occupational Safety and Health Administration (OSHA) guidelines require employers to provide 15-minute breaks for each hour when the temperature exceeds 26 C. However, loss of working hours results in longer workdays.⁴ Use of Sun-Safety Behaviors (upstream): Despite the high intensity of labor in agriculture, there is limited knowledge about the use of sun-safety behaviors. In a study conducted through questionnaires among outdoor workers, they noted that a common barrier was forgetting (49.2%) due to labor, and the lack of built-in sun protection (59.0%).⁵ Workplace safety regulations (downstream): Under OSHA, workers performing high-intensity work in temperatures exceeding 26 C are provided with a 15-minute break every hour. These breaks are extended by 2 C for every 2 C increase, which reduces the workload and intensity burden on workers.^{1,6}	High Labor intensity is of high importance because it directly affects adverse heat exposure outcomes, including heat stress and dehydration.¹⁻³ Additionally, these determinants are linked to other determinants, including quality of work conditions, use of preventive behaviors, and implementation of workplace safety regulations.^{1,4,5} For example, several studies showed that workers performing strenuous tasks had a greater decline in kidney function over the summer harvest months ($p < 0.001$).^{7,8} Although this determinant operates at the individual level, sustained changes can be made at the community or societal levels through national workers' safety regulations, such as OSHA's required break times.⁶	Medium The level of labor intensity shows moderate changeability because interventions focus on changing workplace conditions, such as implementing sun-safety educational programs, longer breaks, and water access, rather than reducing labor intensity.⁹ For example, a study piloting a hydration backpack found that 90% of workers confirmed its reliability and 53% reported using this intervention regularly.^{10,11} This intervention lowers the risk of dehydration during strenuous labor and is relatively inexpensive to produce, with each backpack costing \$20.¹¹ A study conducted in North Carolina reported that farm workers spent between 2 and 7.5 hours working in conditions exceeding recommended limits, resulting in 100% of workers being dehydrated.¹⁰ Implementation of labor laws affects a larger population, such as adding longer, more frequent breaks during higher temperatures. An example of this policy change is OSHA's regulations. As the temperature increases, OSHA requires more break times for outdoor workers, especially vulnerable populations such as farm workers who have an average fatality rate 20 times higher than the US civilian worker per year.^{6,12} An intervention to implement water and rest breaks showed a significant reduction in the prevalence of heat-stress symptoms, including excessive thirst, sweating, and tiredness (76% reduced to 24%).¹³

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Quality of Workplace Conditions (Breaks, Water Access) Quality of workplace conditions encompasses employer-provided heat-related safety precautions, including scheduled rest breaks and accessible drinking water at outdoor agricultural worksites.¹⁴ This determinant is measured by the frequency of breaks and the proximity of clean drinking water, mostly by surveying workers. Inadequate employer provision drives dehydration and heat-related illness among rural Southern farmworkers.¹⁴	Neighborhood/Community Level This determinant operates at the neighborhood/community level. The employer controls break schedules and water provision, not the individual worker nor the federal government.¹⁴ Studies show that voluntary worker behavior was insufficient, even though workers shared that breaks and water should be prioritized by employers.¹⁵	Regulatory Policies (UPSTREAM): When enforcement is weak or absent, basic protections fail to manifest. In California, where heat illness prevention standards exist, only 74% of agricultural employers were found to comply.¹⁶ Many farmworkers lack sufficient regulation of heat illness prevention measures, reinforcing that policy is upstream of workplace condition quality.¹⁷ Level of Labor Intensity (DOWNSTREAM): Poor workplace conditions increase physiological strain and sustain high labor intensity. A Florida study discovered that agricultural workers had only two daily breaks on average, and 66% finished the workday dehydrated.¹⁴ Without adequate rest, workers cannot recover between exertion, reaching dangerous levels of labor intensity. Incidence of HEIs (DOWNSTREAM): Inadequate conditions directly drive heat illness. Over a third of North Carolina Latino farmworkers reported heat illness while working outside.¹⁸ A different study further documented that 11% of study participants experienced one or more serious heat-related illness symptoms across data collection days, directly linking poor break and hydration conditions to HEI incidence.¹⁴ Quality of Workplace Conditions is closely linked to 3 different determinants. Even modest improvements in break frequency and water access could produce compounding reduction in heat illness, making it a high-target leverage point in the system.	High Quality of workplace conditions has high importance due to its substantial impact on the health issue, presence in literature, and operates at the higher community SEF level. Workers who averaged only two daily breaks and drank about a third of recommended beverage consumption had 66% dehydration rates by the end of their shift.¹⁴ Additionally, an integrative review found that multiple studies conducted across Georgia and the Carolinas emphasize access to water, shade and breaks.¹⁷ This determinant is also supported by evidence from multiple study types, including quantitative and qualitative surveys and epidemiological data.^{14,15,18}	Medium Quality of workplace conditions has a medium changeability due to logistically feasible intervention but challenging compliance. In one study, California's rules providing outdoor workers access to shade water, and rest saved lives, but only after the state clarified that water had to be drinkable and free. In this case, deaths and heat-related illnesses still occurred – nearly 8% of workers were still at risk of HRI even on compliant farms, revealing that policy alone cannot fully eliminate risk of illness.¹⁶ It is important to note that while California evidence is most abundant, there are stronger labor protections, larger farmworker populations, and more infrastructure than the rural South, with no equivalent standards currently in rural Southern states. However, studies have shown that improved access to rest breaks can be effective in reducing heat-related health outcomes with minimal impacts on food production.¹ Several studies found that farmworkers lack access prevention measures such as water, shade, and breaks, and that few employers were implementing OSHA's guidelines, reducing the feasibility of change.¹⁷ Furthermore, change would require policy-level intervention and defy the Southern states' active resistance to heat standard legislation, maintaining this determinant's medium status.¹⁹
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Immigration Status Immigration status is defined as the legal classification of an individual's residency and work allowance in the US. Farm workers are primarily divided into two immigration statuses, those being H2A employees, who have temporary legal status to complete agricultural labor and non-H2 employees,²⁰ which includes US citizens, green card holders, those with other visa types, and undocumented workers.	Country and Society This determinant operates at the Country and Society level of the SEF framework. An individual's immigration status is defined solely by federal regulations, and an individual's status can be reclassified based on changes at the federal level.²¹ At the state level, states have partial discretion over policies that determine the rights and public program eligibility of noncitizens.²¹	Use of Sun-safety Behaviors (DOWNSTREAM): H-2A workers implement the use of sun safety strategies at a lower rate than their non-H-2A counterparts. In a 2010 survey, adaptive strategies to protect against heat-related illnesses were reported less frequently among H-2A workers.²² Regulatory Policies (UPSTREAM): Regulatory policies on immigration lead to an increase in H-2A workers, as it is a legal form of immigration. For example, the American Immigration Council poses that increased regulatory policies at the state level have significantly contributed to an increase in H-2A status among agricultural workers.²³ Annual Wage Income (DOWNSTREAM): H-2A status leads to increased annual wage income due to legal regulations Immigrants admitted through the H-2A program enjoy higher wages.²⁴ This is due to US Department of Labor restrictions on deceptive payment structures like unauthorized deductions.²⁵	Medium importance Non-H-2A farm workers displayed significantly higher knowledge of heat related illness than H-2A workers,²⁶ as well as more adaptive strategies against it. However, it is important to note that while adaptive strategies were reported less frequently among H-2A workers, symptoms of heat illness were also less frequently reported.²² While this may be due to inaccurate reporting by subjects, it shows that there may not be a clear delineation between H-2A and non-H-2A workers' health outcomes, granting this determinant medium level importance despite its high SEF level.	Medium changeability We observe that the demand for H-2A workers has risen by 35.2 percent between 2017 and 2022, reflecting an increased demand for agricultural labor.²⁷ However, those who are critics of immigration reform believe that guest worker programs provide an avenue for immigrants to enter a country legally, then overstay their visa and become undocumented.²⁴ This has led to crackdown programs like the Secure the Border Act, of 2023, which "required the Secretaries of Homeland Security and Agriculture to submit a report on the number of people in the agricultural workforce and the number of foreign nationals in the agricultural workforce who are authorized for this work (H-2A workers)".²⁸ While there is clear motivation and reason to change the framework around immigration for agricultural labor, there are political roadblocks present as well, placing the changeability at a medium level.
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Annual Wage Income	Country and Society	Higher H-2A and undocumented worker status is associated with lower wages (DOWNSTREAM):	High Importance	Low Changeability
The National Agricultural Workers Survey (NAWS) from 2021–2022 found, when adjusted for family size poverty lines, 22.7% of farmers have household incomes below the federal poverty line.²⁹ Annual wage income is operationalized as annual household earnings at or below the federal poverty level (FPL). At the same time, research suggests that more generous minimum wage policies may help reduce food insecurity among low-income households.³⁰	Annual wage income operates at Country and Society level. Suppressed agricultural wages stem from longstanding Fair Labor Standards Act exemptions,³¹ while organizations like Farmworker Justice advocate for change through political advocacy to strengthen labor protections and enforcement.³² Together, policy and institutional forces set the parameters within which household income outcomes are determined.	The unadjusted hourly wage gap between undocumented workers and native workers is about 40%, and even after controlling for education and other factors, undocumented workers continue to face a penalty in income of about 20%.³³ Additionally workers with H-2A status are in weaker positions to bargain for wages due to their lack of permanent residency in the U.S.. Lower wages also reduce sun-safety behavior adoption (DOWNSTREAM): Workers may reject sun safety equipment due to productivity issues (piece-production) when their employer does not provide it, and the lack of overtime regulation of work for farmers under the Fair Labor Standards Act strengthens this effect.³¹ Research on NC farmworkers shows that pesticide safety violations and wage violations often occur together, suggesting that lower wages are tied to poorer overall working conditions.³⁴ Lastly, research shows that workers with piece/production rates are 6.20 as likely to (OR: 6.20; 95% CI: 1.11–34.54) experience heat related illnesses compared to those who are paid through other methods.¹ There are 2 relationships from and to Annual Wage Income.	Annual household income is a high-importance determinant because annual wage income operates at the socioeconomic level of the Social-Ecological Framework, exerting broad upstream influence due to policy being the largest influencer.³⁵ Farmworkers are 20 times more likely to die from heat exposure than other workers, and low wages further limit their ability to cope with and recover from heat-related illness.³⁶ Additionally, a literature review found that farmers who are paid by piece/production rates are 6.20 as likely to (OR: 6.20; 95% CI: 1.11–34.54) experience heat related illnesses compared to those who are paid through other methods.¹	Annual wage income among Southern agricultural workers is difficult to change meaningfully through policy due to structural, legal, and political barriers unique to this population and region. Some wage improvement has occurred nationally; real wages for non-supervisory farmworkers rose moderately over the past decade, although farmworkers still earned about 60% of comparable non-farm workers as of 2022.³⁷ Attempting to change wages has had minimal success through consumer demand bonuses through retailers, as shown through the Coalition of Immokalee Workers' "Fair Food Premium", which resulted in farmworkers earning over \$20 million in bonuses between January 2011 and October 2015, but has not been widely adopted outside of Florida.³⁸ 2017 NAWS research shows that, among agriculture workers, union members earn \$0.87 more in hourly wages and are 4.8% more likely to receive a bonus.³⁹ While individual states can enact minimum wage laws more inclusive than the FLSA, several exemptions mean many Southern farmworkers remain uncovered in practice.³¹ Meaningful change would require state or federal legislative action, facing significant industry opposition and a lack of bipartisan support.

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Use of Sun-safety Behaviors (Including HRI Prevention Training) Sun-safety behaviors (hydration, shade breaks, appropriate clothing) and HRI training (symptom recognition and response) are proactive strategies for protecting agricultural workers from extreme heat.^{17,40} These are measured in research via self-reported behavioral surveys and knowledge assessments,^{26,40} as well as in workplaces through enforced administrative controls, hydration provisions, and mandatory safety programs.¹⁷	Individual Level This determinant primarily operates at the individual level because it is dictated by the worker's behaviors, knowledge, and training exposure.^{26,40,41} These measures capture personal actions that directly shape immediate heat risk during work, even though workplace conditions can enable or constrain them.^{15,42}	Level of Workforce Intensity (Upstream): The Food and Agriculture Organization (FAO) of the United Nations has identified several work-related factors in which increase the risk of heat exposure, whilst simultaneously decreasing an individual's likelihood of using sun-safety behaviors.⁴³ Of which, the literature notes that farm managers or supervisors tend to rely on the workers to opt into sun-safety behaviors but are doing so at the risk of losing pay.^{1,15} Despite this, the FAO states that the best interest of employers is to promote sun-safety behaviors, as it increases labor productivity.⁴³ Prevalence of H-2A Status Among Workers (Upstream): Studies have shown that H-2A workers have significant differences in heat safety knowledge, compared to their non H-2A colleagues ($p < 0.001$).²⁶ Annual Wage Income (Upstream): Qualitative evidence has provided in several studies that many individuals who work in agriculture live in low socioeconomic conditions, and their low wages hinder their abilities to receive basic resources.¹ Additionally, another study found that workers consistently attributed financial reasons as the most "influential" reasoning behind why they prefer not to spend money on sun-safety items, specifically clothing.⁴² In all, there are 3 relationships leading to the Use of Sun-Safety Behaviors – It is the most downstream determinant yet linked with many others.	Low Importance This determinant is of lower importance because sun-safety behaviors and HRI-prevention training can meaningfully reduce heat risk, but their effectiveness depends on workers consistently applying them in real situations.^{26,40} Its relationship to heat exposure and HRIs is supported across multiple study types, including cross-sectional, qualitative, intervention, and review evidence.^{1,15,17,41} Still, because it operates mainly at the individual SEF level and is often enabled or constrained by employers and work conditions, its importance is best rated as moderate.⁴²	Moderate Changeability This determinant appears moderately changeable because tested interventions show that heat-safety knowledge can improve, but evidence for sustained behavior or outcome change is more limited. In a comparison-group intervention, participatory heat education increased mean knowledge scores from 4.6 to 6.3 in the intervention group, with greater improvement than the comparison group (mean change 1.6 vs. 0.41, $p = 0.04$).⁴¹ Other studies also suggest feasible points of change through OSHA-based training, mobile heat-safety tools, and acceptable personal cooling interventions.^{15,26} Reviews further support that training, hydration, shade, rest, and protective practices are modifiable prevention targets, though implementation is often inconsistent and constrained by worksite culture, access, cost, and weak enforcement.^{1,17,42} Since changing this determinant usually requires action at both the individual and employer or jobsite levels, also with considerable resource needs, its overall changeability is not very high but is also relatively feasible with the right interventions. For these reasons, its changeability is in the intermediate-moderate level medium.
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Summary:

In this table, five key determinants of heat exposure illnesses (HEIs) among agricultural workers in the rural, Southern United States are examined: level of labor intensity, quality of workplace conditions, immigration status, annual wage income, and use of sun-safety behaviors. All leverage points were evaluated considering SEF level, system location, importance, and changeability.

Level of workforce intensity operates at the individual level and is rated as highly important but has medium changeability, as interventions target workplace conditions rather than reducing workload itself. Quality of workplace conditions functions at the community level, with high importance and medium changeability; modest improvements in break frequency and water access could meaningfully reduce HEI incidence. Immigration status operates at the country/society level with medium importance and changeability, as political barriers complicate reform. Annual wage income, also at the country/society level, carries high importance but low changeability due to deep structural and legislative barriers. Sun-safety behaviors operate at the individual level with moderate importance and changeability, contingent on employer support and resource access.

The determinant that acts most as a leverage point is **Quality of Workplace Conditions**. It operates at the neighborhood/community level, a higher tier of the SEF, making it more actionable than society-level determinants like wage income or immigration status, yet broader in reach than individual behaviors. Its importance is high, with direct evidence linking inadequate breaks and water access to dehydration and HEIs across multiple study types.^{14,15,44} Critically, it intersects with multiple other determinants. For example, poor conditions worsen labor intensity, constrain sun-safety behavior adoption, and reflect failures in regulatory policy. Lower-cost and targeted improvements like mandating accessible drinking water and structured rest breaks could produce compounding benefits across the system with feasible implementation, making it the strongest leverage point for reducing HEIs with currently available resources.

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AI Attestation: We attest that this submission did not use AI in its development or in the creation of any of its components.